

NB15: Panel-Bazli Egitim & Stacking (v2)

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Deney Tasarimi

Her panel (CFTR/KANSER/PAH) icin ayri model. Dort senaryo x dort model tipi (LightGBM, CatBoost, NN, DNN). Birincil metrik: pathogenic F1. S1: panel %50 + dengelenmis MASTER cekirdek (625/625). S2: panel %50 + ayni boyutta MASTER (2x). S3/S4: NN/DNN pretrain(MASTER)->finetune(panel). Test'te MASTER YOK.

KRITIK: Final Test Dagilimi (%80 benign / %20 patho)

Final test ~%80 benign (train'in TERSI). Threshold uc modda (f1_raw/f1_8020/mcc_8020), test %50/50 + bootstrap %80/20 (N=50, %95 CI). Birincil: %80/20 F1.

NN/DNN Iyilestirme

Agac: 5-fold CV grid. NN/DNN: 5-fold CV + FocalLoss(gamma=2) + class weighting + early stopping + SmallMLP (yuksek dropout, weight_decay).

Missing & Veri Butunlugu

M3: >%50 NaN -> is_missing flag + medyan impute. Birebir-ayni satir drop: CFTR=0, KANSER=3, PAH=3. 63 sutun drop -> 288 feature.

1. Sonuclar (f1_8020 threshold) - %50/50 vs %80/20 bootstrap

Scn	Panel	Model	Thr	Te-F1(50/50)	Te-P	Te-R	F1(80/20)	80/20 CI
S1	CFTR	lightgbm	0.55	0.861	1.000	0.756	0.806	[0.11-1.00]
S1	CFTR	catboost	0.50	0.875	1.000	0.778	0.850	[0.50-1.00]
S1	CFTR	nn	0.50	0.685	0.893	0.556	0.453	[0.06-0.67]
S1	CFTR	dnn	0.51	0.618	0.913	0.467	0.422	[0.00-0.75]
S2	CFTR	lightgbm	0.38	0.899	0.909	0.889	0.542	[0.25-0.60]
S2	CFTR	catboost	0.42	0.864	0.884	0.844	0.477	[0.22-0.55]
S2	CFTR	nn	0.51	0.871	0.925	0.822	0.565	[0.29-0.67]
S2	CFTR	dnn	0.51	0.884	0.927	0.844	0.598	[0.33-0.67]
S3	CFTR	nn	0.27	0.747	0.933	0.622	0.515	[0.33-0.75]
S4	CFTR	dnn	0.38	0.779	0.938	0.667	0.526	[0.00-0.75]
S1	KANSER	lightgbm	0.37	0.911	0.898	0.925	0.651	[0.62-0.68]
S1	KANSER	catboost	0.72	0.848	0.936	0.774	0.682	[0.53-0.78]
S1	KANSER	nn	0.51	0.790	0.895	0.707	0.560	[0.39-0.67]
S1	KANSER	dnn	0.54	0.774	0.918	0.669	0.598	[0.43-0.72]
S2	KANSER	lightgbm	0.36	0.910	0.875	0.947	0.600	[0.57-0.62]
S2	KANSER	catboost	0.66	0.839	0.904	0.782	0.629	[0.51-0.72]
S2	KANSER	nn	0.65	0.757	0.873	0.669	0.524	[0.40-0.63]
S2	KANSER	dnn	0.60	0.692	0.936	0.549	0.563	[0.35-0.74]
S3	KANSER	nn	0.47	0.817	0.916	0.737	0.622	[0.50-0.74]
S4	KANSER	dnn	0.53	0.798	0.958	0.684	0.690	[0.48-0.81]
S1	PAH	lightgbm	0.45	0.917	0.894	0.941	0.453	[0.33-0.48]
S1	PAH	catboost	0.60	0.908	0.908	0.908	0.489	[0.33-0.53]
S1	PAH	nn	0.48	0.854	0.887	0.824	0.418	[0.29-0.50]
S1	PAH	dnn	0.48	0.846	0.886	0.810	0.417	[0.29-0.50]
S2	PAH	lightgbm	0.22	0.913	0.844	0.993	0.360	[0.33-0.36]
S2	PAH	catboost	0.51	0.903	0.897	0.908	0.454	[0.36-0.50]
S2	PAH	nn	0.58	0.814	0.861	0.771	0.377	[0.21-0.46]
S2	PAH	dnn	0.66	0.751	0.871	0.660	0.362	[0.18-0.47]
S3	PAH	nn	0.30	0.885	0.863	0.908	0.385	[0.30-0.42]
S4	PAH	dnn	0.54	0.655	0.918	0.510	0.405	[0.24-0.57]

1b. Train Sonuclari (f1_8020 thr) + Train-Test F1 farki

Scn	Panel	Model	Tr-F1	Tr-P	Tr-R	Tr-MCC	Te-F1(50/50)	Tr-Te farki
S1	CFTR	lightgbm	0.964	0.981	0.948	0.929	0.861	+0.104
S1	CFTR	catboost	0.931	0.934	0.928	0.859	0.875	+0.056
S1	CFTR	nn	0.841	0.898	0.791	0.700	0.685	+0.156
S1	CFTR	dnn	0.789	0.900	0.701	0.632	0.618	+0.171
S2	CFTR	lightgbm	0.989	1.000	0.978	0.943	0.899	+0.090
S2	CFTR	catboost	0.989	1.000	0.978	0.943	0.864	+0.125

S2	CFTR	nn	0.955	0.966	0.944	0.766	0.871	+0.084
S2	CFTR	dnn	0.967	0.967	0.967	0.817	0.884	+0.083
S3	CFTR	nn	0.850	0.971	0.756	0.526	0.747	+0.103
S4	CFTR	dnn	0.878	0.973	0.800	0.575	0.779	+0.099
S1	KANSER	lightgbm	0.987	1.000	0.974	0.973	0.911	+0.076
S1	KANSER	catboost	0.928	0.975	0.885	0.860	0.848	+0.080
S1	KANSER	nn	0.858	0.910	0.811	0.723	0.790	+0.068
S1	KANSER	dnn	0.837	0.918	0.769	0.697	0.774	+0.063
S2	KANSER	lightgbm	0.990	1.000	0.981	0.970	0.910	+0.081
S2	KANSER	catboost	0.979	1.000	0.958	0.937	0.839	+0.140
S2	KANSER	nn	0.825	0.979	0.712	0.629	0.757	+0.067
S2	KANSER	dnn	0.688	0.979	0.530	0.484	0.692	-0.004
S3	KANSER	nn	0.850	0.944	0.773	0.629	0.817	+0.033
S4	KANSER	dnn	0.802	0.958	0.689	0.577	0.798	+0.004
S1	PAH	lightgbm	0.986	1.000	0.972	0.970	0.917	+0.068
S1	PAH	catboost	0.941	0.953	0.929	0.873	0.908	+0.032
S1	PAH	nn	0.841	0.887	0.800	0.677	0.854	-0.013
S1	PAH	dnn	0.796	0.878	0.728	0.608	0.846	-0.051
S2	PAH	lightgbm	0.993	1.000	0.987	0.963	0.913	+0.081
S2	PAH	catboost	0.963	0.993	0.935	0.819	0.903	+0.061
S2	PAH	nn	0.897	0.970	0.834	0.584	0.814	+0.083
S2	PAH	dnn	0.836	0.978	0.731	0.500	0.751	+0.086
S3	PAH	nn	0.898	0.858	0.942	0.222	0.885	+0.012
S4	PAH	dnn	0.581	0.904	0.429	0.155	0.655	-0.074

Ortalama train-test F1 farki (overfit gostergesi): 0.065.

2. Threshold Modu Karsilastirmasi (panel-ort. %80/20 F1)

Threshold modu	Ort. %80/20 F1
f1_raw	0.5137
f1_8020	0.5331
mcc_8020	0.5335

3. Her Panel Icin En Iyi (%80/20 bootstrap F1)

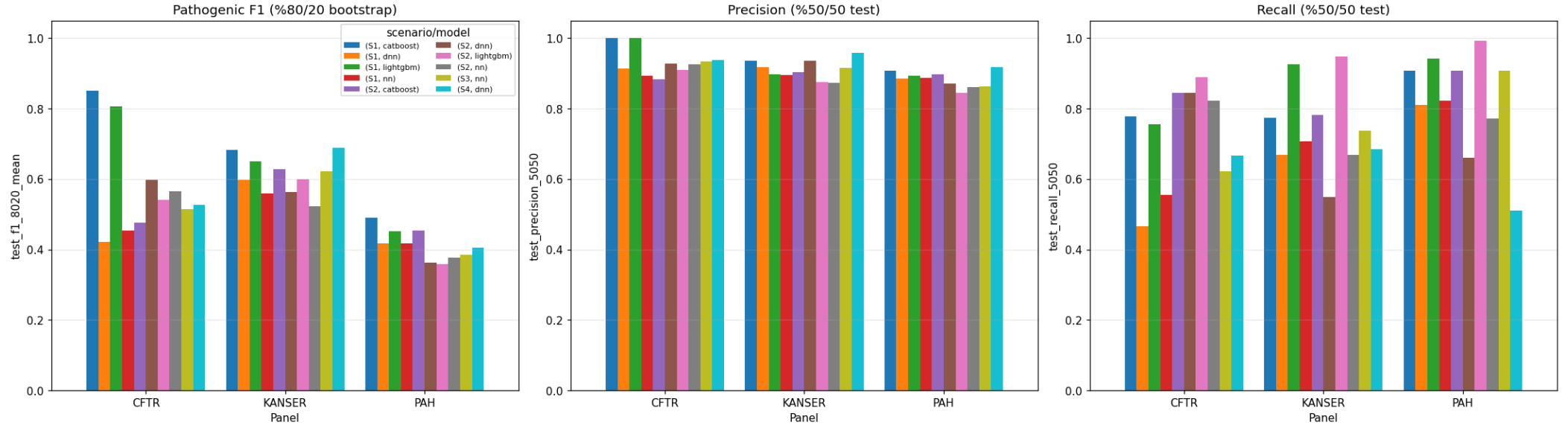
Panel	En iyi	F1(80/20)	80/20 CI	F1(50/50)
CFTR	S1/catboost	0.8500	[0.50-1.00]	0.875
KANSER	S4/dnn	0.6897	[0.48-0.81]	0.798
PAH	S1/catboost	0.4895	[0.33-0.53]	0.908

3b. Secilen NN/DNN Hiperparametreleri

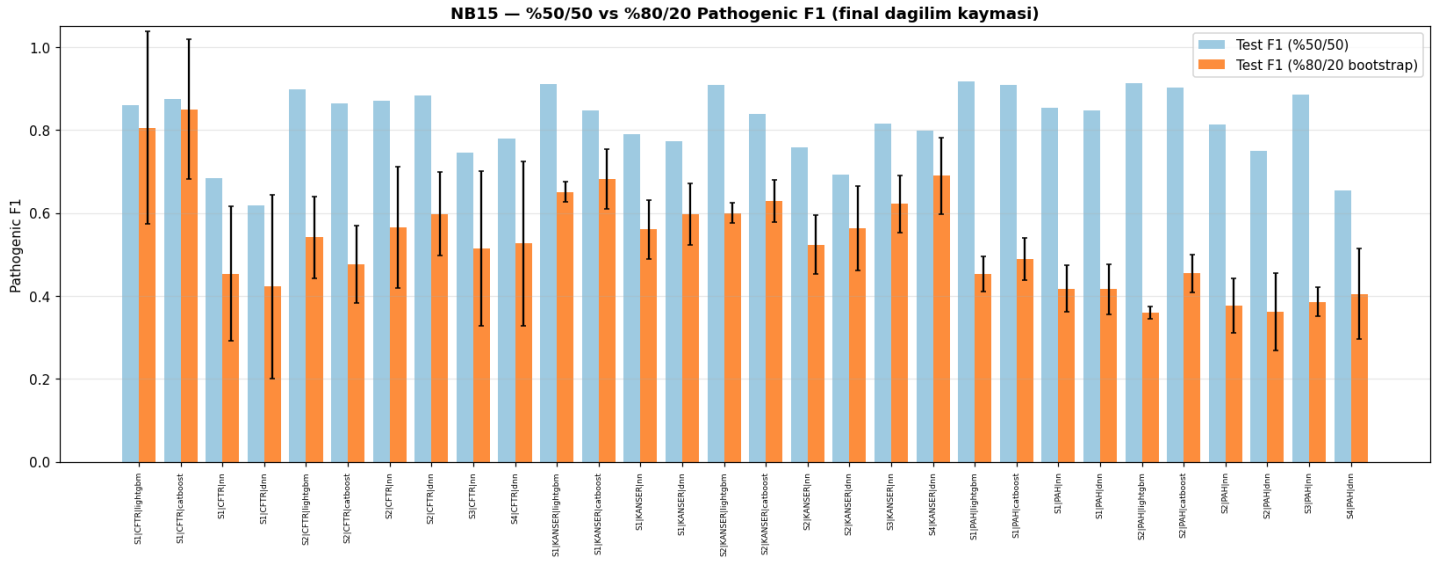
Scn	Panel	Model	Secilen combo
S1	CFTR	nn	{'hidden': 64, 'dropout': 0.3, 'weight_decay': 0.001, 'lr': 0.001, 'focal_gamma': 2.0}
S1	CFTR	dnn	{'hidden': 128, 'n_layers': 3, 'dropout': 0.4, 'weight_decay': 0.001, 'lr': 0.001, 'focal_gamma': 2.0}
S2	CFTR	nn	{'hidden': 128, 'dropout': 0.5, 'weight_decay': 0.001, 'lr': 0.001, 'focal_gamma': 2.0}
S2	CFTR	dnn	{'hidden': 128, 'n_layers': 2, 'dropout': 0.5, 'weight_decay': 0.001, 'lr': 0.001, 'focal_gamma': 2.0}
S3	CFTR	nn	{'hidden': 128, 'dropout': 0.5, 'weight_decay': 0.001, 'lr': 0.001, 'focal_gamma': 2.0}
S4	CFTR	dnn	{'hidden': 128, 'n_layers': 2, 'dropout': 0.4, 'weight_decay': 0.001, 'lr': 0.001, 'focal_gamma': 2.0}
S1	KANSER	nn	{'hidden': 128, 'dropout': 0.3, 'weight_decay': 0.001, 'lr': 0.001, 'focal_gamma': 2.0}
S1	KANSER	dnn	{'hidden': 128, 'n_layers': 2, 'dropout': 0.4, 'weight_decay': 0.001, 'lr': 0.001, 'focal_gamma': 2.0}
S2	KANSER	nn	{'hidden': 128, 'dropout': 0.3, 'weight_decay': 0.001, 'lr': 0.001, 'focal_gamma': 2.0}
S2	KANSER	dnn	{'hidden': 128, 'n_layers': 2, 'dropout': 0.4, 'weight_decay': 0.001, 'lr': 0.001, 'focal_gamma': 2.0}
S3	KANSER	nn	{'hidden': 64, 'dropout': 0.5, 'weight_decay': 0.001, 'lr': 0.001, 'focal_gamma': 2.0}
S4	KANSER	dnn	{'hidden': 128, 'n_layers': 3, 'dropout': 0.5, 'weight_decay': 0.001, 'lr': 0.001, 'focal_gamma': 2.0}
S1	PAH	nn	{'hidden': 64, 'dropout': 0.3, 'weight_decay': 0.001, 'lr': 0.001, 'focal_gamma': 2.0}
S1	PAH	dnn	{'hidden': 128, 'n_layers': 2, 'dropout': 0.4, 'weight_decay': 0.001, 'lr': 0.001, 'focal_gamma': 2.0}
S2	PAH	nn	{'hidden': 128, 'dropout': 0.3, 'weight_decay': 0.001, 'lr': 0.001, 'focal_gamma': 2.0}
S2	PAH	dnn	{'hidden': 128, 'n_layers': 3, 'dropout': 0.4, 'weight_decay': 0.001, 'lr': 0.001, 'focal_gamma': 2.0}
S3	PAH	nn	{'hidden': 128, 'dropout': 0.3, 'weight_decay': 0.001, 'lr': 0.001, 'focal_gamma': 2.0}
S4	PAH	dnn	{'hidden': 128, 'n_layers': 2, 'dropout': 0.4, 'weight_decay': 0.001, 'lr': 0.001, 'focal_gamma': 2.0}

4. Metrikler (%80/20 F1, Precision, Recall)

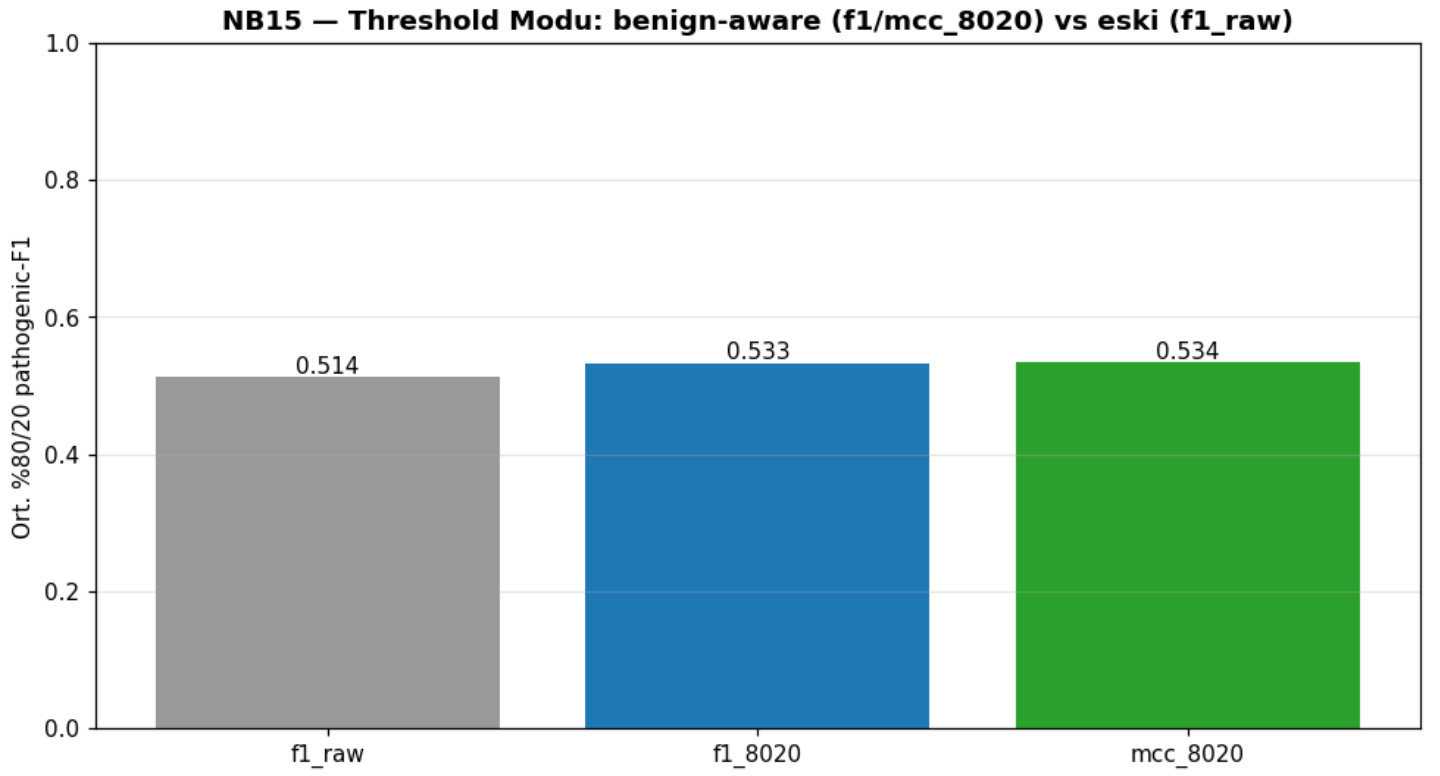
NB15 — Panel-Bazli Metrikler (birincil: %80/20 pathogenic-F1)



5. %50/50 vs %80/20 F1

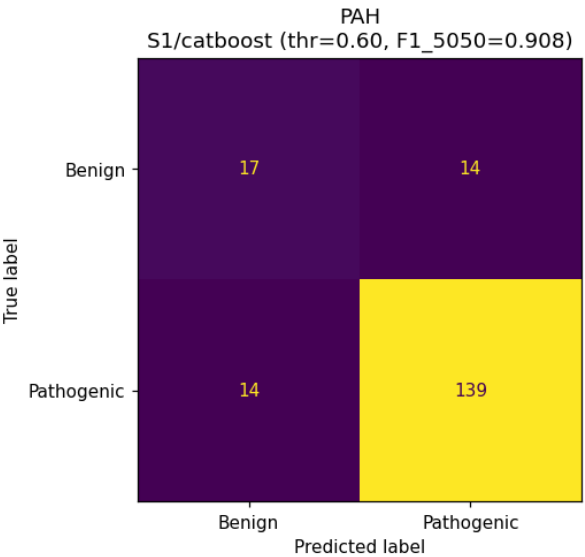
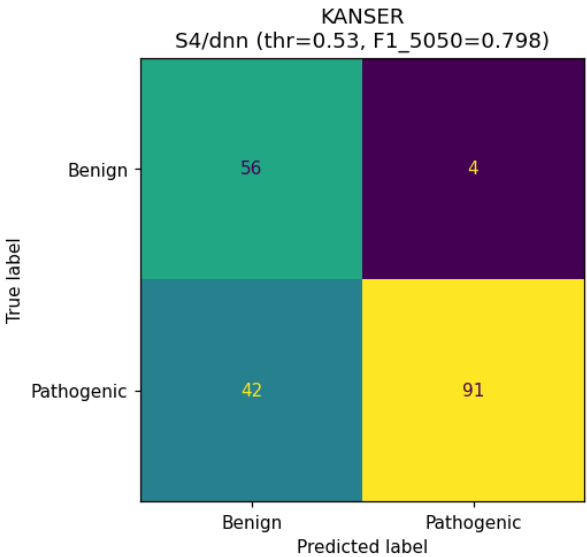
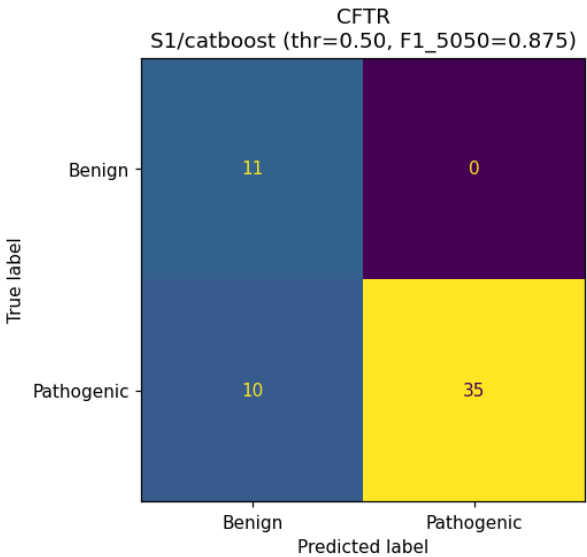


6. Threshold Modu Karsilastirmasi



7. Confusion Matrix'ler

NB15 — Confusion Matrices (panel basina en iyi, %50/50 test)



8. Feature Importance

NB15 — Top-15 Feature Importance (panel basina en iyi agac modeli)

